RAMAN SPECTROSCOPY

□ State the criteria for a molecule to be Raman active. Give at least three examples of molecules which can give Raman spectra.

Ans:

In order to be Raman active, the molecular rotation or vibration must cause some change in molecular polarizability. The change of polarizability may be either magnitude or the direction of the polarizability ellipsoid. Hence all the vibrational modes of a molecule are not necessarily Raman active as they may not cause a change in polarizability and therefore will not appear in a Raman spectrum.

Some Raman Active molecules are: H_2, O_2, H_2O, N_2O, $CHCl_3$ etc.

□ State and explain the rule of mutual exclusion with example.

Ans:

The rule of mutual exclusion states - If a molecule has a center of symmetry then RaMan active vibrations are infra-red inactive and vice versa. If there is no center of symmetry then some vibrations may be both Raman & infra-red active.

Example: CO_2 has a center of symmetry. It's symmetic stretching is Raman active but infra-red inactive. On the other hand asymmetric stretching and bending modes

are infra-red active but both are Raman active. Whereas, N_2O molecule having no center of symmetry shows both Raman and infra-red spectrum.

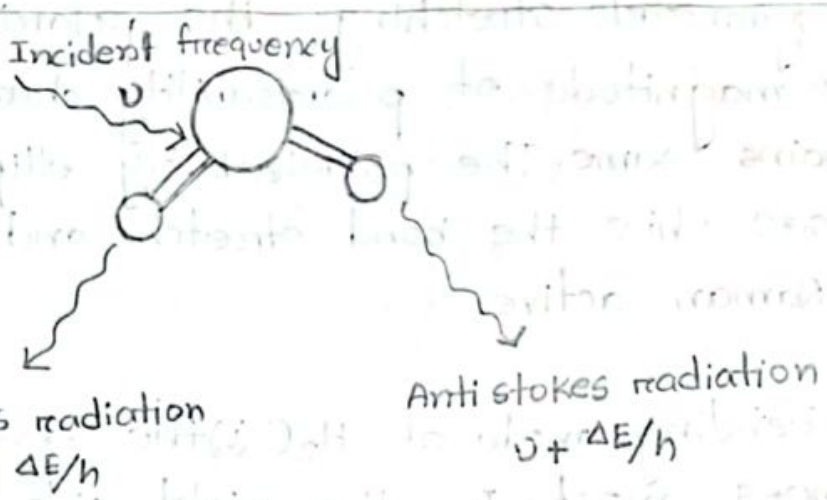
□ According to the quantum theory of Raman spectra explain stokes and anti-stokes radiation. Which radiation has a higher frequency ? / The Quantum Theory of Raman effect.

Ans:

Stokes radiation: Radiation scattered with a frequency lower than that of incident beam is called stokes' radiation. In an elastic scattering, let the exchange of energy between a molecule and photon is ΔE Joules. ΔE must represent a change in rotational or vibrational change in the molecule. If molecule gains energy, ΔE the photon will be scattered with energy $h\upsilon - \Delta E$ and the equivalent radiation will have a frequency $\upsilon - \Delta E/h$ and the less radiation of photon is known as stokes' radiation.

Anti stokes radiation: Radiation scattered with frequency higher than that of incident beam is called anti-stokes radiation. In an elastic scattering, if a molecule loses energy ΔE Joules, the photon will be scattered with energy $h\upsilon + \Delta E$ and the equivalent radiation will have a frequency $\upsilon + \Delta E/h$. Thus, anti-stokes radiation has a higher frequency than stokes' radiation.

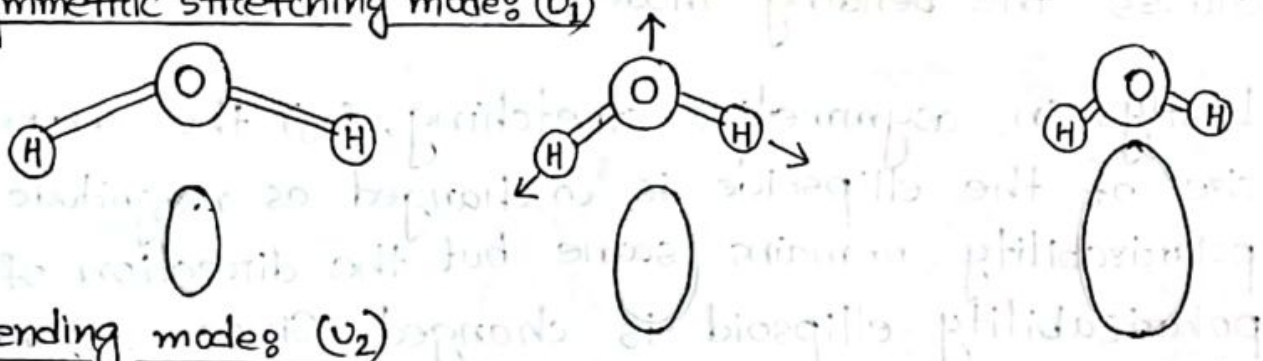
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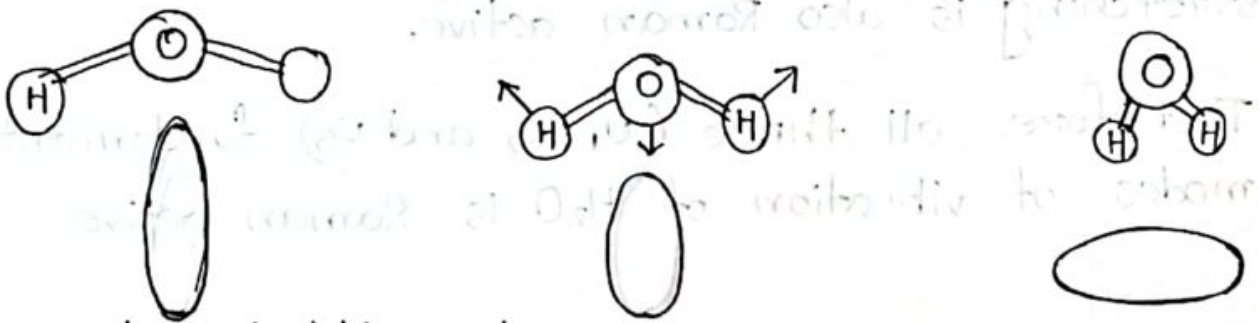
□ Explain the Raman activity of the fundamental modes of vibration in the H_2O molecule.

Ans:

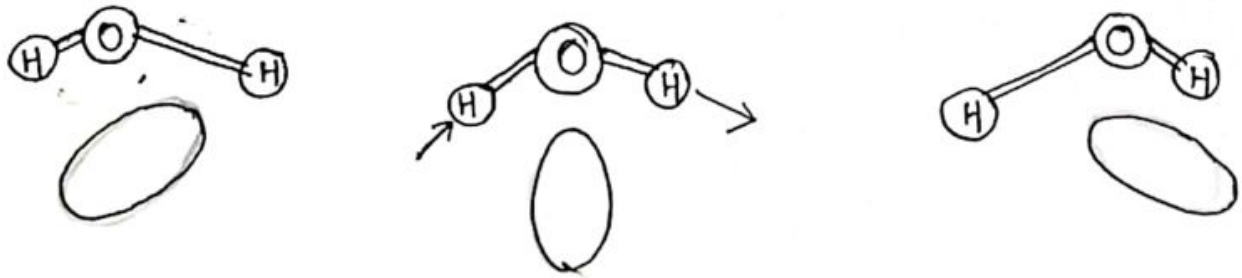
Symmetric stretching mode: (υ_1)



Bending mode: (υ_2)



Asymmetric stretching mode: (υ_3)



In symmetric stretching, the polarizibility changes, i.e. the magnitude of polarizability changes but direction remains same. The polarizability ellipsoid of H_2O decrease in size while the bond stretch and vise versa. Thus, υ_1 is Raman active.

In bending mode of H_2O (υ_2) the shape of the ellipsoid changes most. In the right side, the molecule approaches with a verticle axis and polarizability changes. Similiarly in the left, the molecule approaches linear with horizontal axis. This changable polarizability causes the bending mode of H_2O to be Raman active.

Lastly, in asymmetric stretching, (υ_3) the shape and size of the ellipsoide is unchanged as magnitude of polarizability remains same but the direction of the polarizability ellipsoid is changed. Since, asymmetric stretching is also Raman active.

Therefore, all three (υ_1, υ_2 and υ_3) fundamental modes of vibration of H_2O is Raman active.

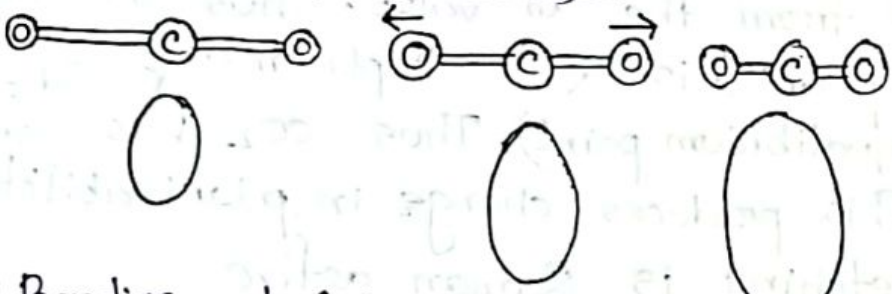
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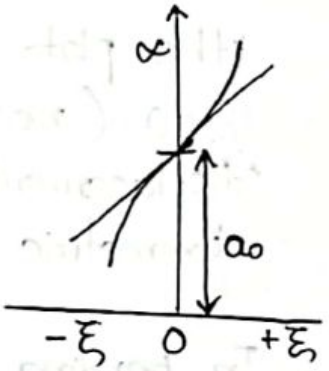
□ Symmetric stretching mode of CO_2 molecule is Raman active whereas bending and asymmetric stretching modes are observed to be Raman inactive. Explain the reasons with the plot of vibration of polarizability with displacement during vibration.

Ans:

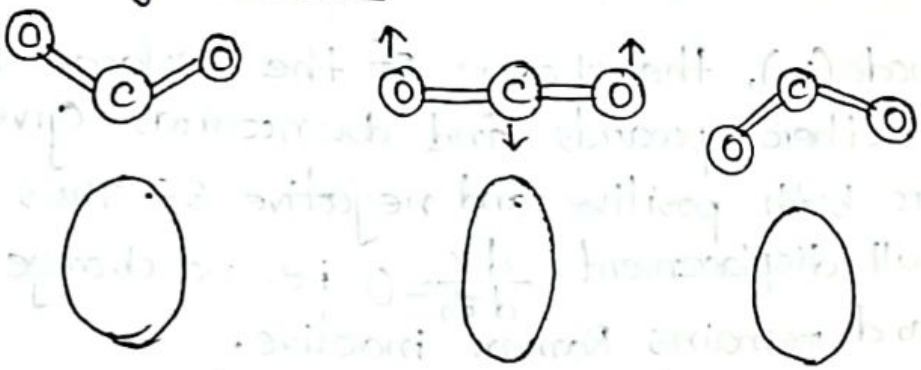
The vibrational modes of CO_2 are -

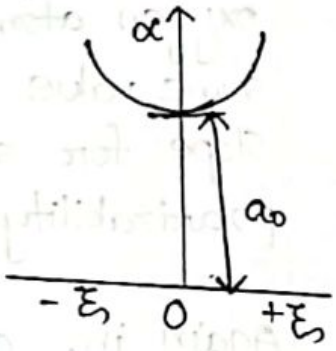
(i) Symmetric stretching mode(υ_1):



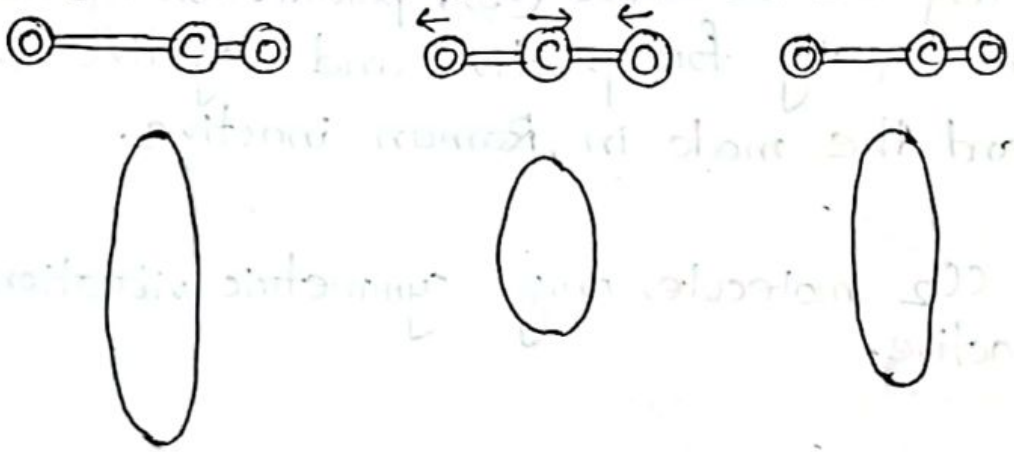


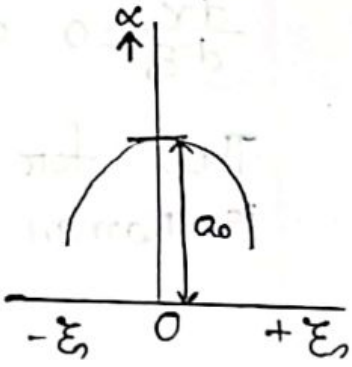
(ii) Bending mode (υ_2):





(iii) Asymmetric stretching mode (υ_3):





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The change of polarizability with some displacement, ($\bar{\varepsilon}$) can be a consideration to describe Raman active or inactive modes of vibration. $+\bar{\varepsilon}$ is a measure of extension and neg. $-\bar{\varepsilon}$ is a measure of compression.

In the υ_1 mode, when the bond of CO_2 is stretched. polarizability value increases from α_0 and when it decreases from the α_0 value. Thus the slope of the plot of polarization with displacement $\frac{d\alpha}{d\bar{E}} \neq$ at $\bar{\varepsilon}=0$ (near equalibrium point). Thus CO_2 has some polarized with displacement. This produces change in polarizability. Thus symmetric stretching is Raman active.

In bending mode (υ_2), the change of the displacement of oxygen atom wetherc upwards and downwards gives the same value for both positive and negative $\bar{\varepsilon}$. Thus the slope for small displacement, $\frac{d\alpha}{d\bar{\varepsilon}}=0$ i.e. no change in the polarizability and remains Raman inactive.

Again in asymmetric stretching mode (υ_3), polarizability changes or decreases equally for positive and negative $\bar{\varepsilon}$. Thus $\frac{d\alpha}{d\bar{\varepsilon}}=0$ and the mode in Raman inactive.

Thus for CO_2 molecule, only symmetric vibration mode is Raman active.

□ Symmetric top molecules are Raman active but spherical top molecules are Raman inactive. Explain with examples.

Ans 8

Symmetric top molecules because of their axial symmetry, have polarizibility ellipsoids similare to those of linear molecules. For a molecule such as chloroform $CHCl_3$, where the chlorine atoms are bulky. The polarizability is greater across the symmetry axis and the minor axis of the ellipsoid lies in the direction. But along the symmetry axis no change of polarization occurs, thus polarizibility ellipsoid remains spherical.

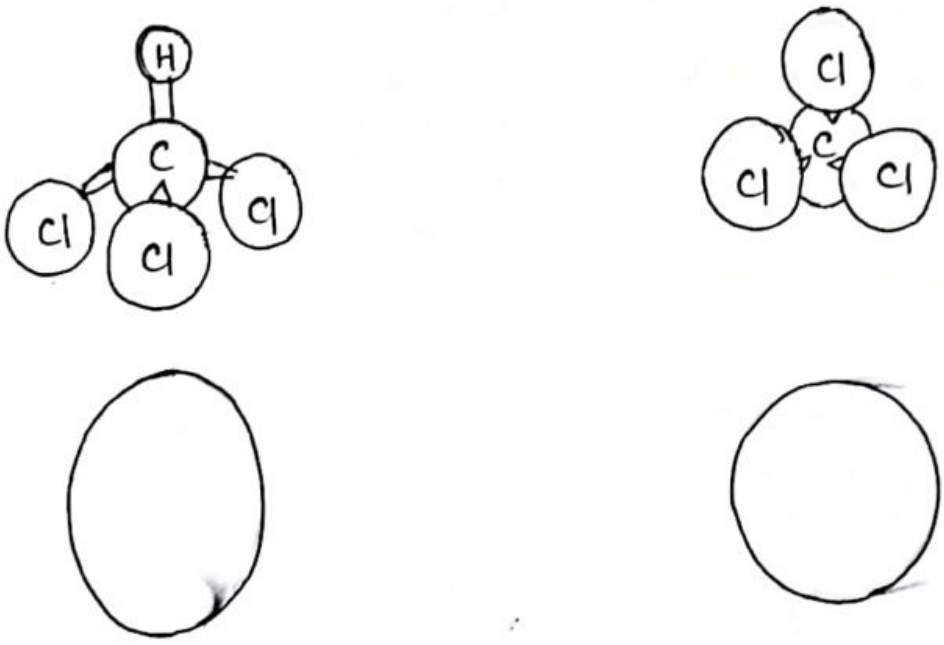


Fig: The chloroform molecule and its polarizability ellipsoid seen from across and along the symmetry axis.

Since, the polarizability of a spherical symmetric top molecule changes along and across the bond axis, symmetric top molecules are Raman active.

Spherical top molecules are those with tetrahedral symmetry such as CH_4 or SiH_4. The polarizability ellipsoid for such molecules is spherical sphere and the rotation of this ellipsoid will produce no change in polarizability. Therefore the pure rotations of spherical top molecules are completely Raman inactive.

□ Define molecular polarizability.

Ans:

Polarizability refers to the tendency of a molecule, when subjected to an electric field to acquire an electric dipole moment in proportion to the applied field. When a molecule placed in a static electric field(E), the induced dipole moment,

$$\mu = \alpha E \quad ; \alpha = \text{Polarizability}.$$

□ Mention some differences between stokes and anti-stoke radiation.

Ans:

Stokes Line/radiation.	Anti-stokes Line/radiation
1. The frequency of scattered radiation is less than of the incident light.	1. The frequency of scattered radiation is more than of the incident light.
2. More intensive with a high intensity of absorption.	2. Less intense with low intensity of absorption.
3. Absorption of vibrational energy by system.	3. Emission of vibrational energy by the system.
4. At low temp, stokes line take place more frequently.	4. At high temp. molecules takes place in anti stokes radiation.

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□ Write energy expression and selection rule of Raman spectra in case of symmetric top molecules. Also show that R branch lines near at $4B, 6B, 8B \cdots cm^{-1}$ from the exciting line.

Ans:

The rotation of a symmetric top molecule about the top axis produces no change in the polarizability but end to end rotations will produce a change in the polarizability.

For a symmetric top molecule the energy levels,

$$\varepsilon_{J,K} = BJ(J+1) + (A-B)K^2 \ cm^{-1}$$

$$(J = 0, 1, 2 \cdots ; K = \pm J, \pm (J-1), \cdots)$$

The selection rules for Raman spectra are:

$$\Delta K = 0$$

$$\Delta J = 0, \pm 1, \pm 2 \quad (\text{except for } K=0 \text{ states when } \Delta J = \pm 2 \text{ only})$$

$\Delta K = 0$ implies that changes in angular momentum about the top axis will give no rise to Raman spectra and remains Raman inactive.

When $\Delta J = +1$, R branch lines are obtained,

$$\Delta \varepsilon_R = \varepsilon_{J+1} - \varepsilon_J$$

$$= B(J+1)(J+1+1) - BJ(J+1)$$

$$= B(J+2)(J+1) - BJ(J+1)$$

$$= B\{J^2 + 2J + 1 + 2 - J^2 - J\} \quad = B(2J+2)$$

$$= 2B(J+1) \ cm^{-1}$$

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$\therefore \Delta \bar{\nu}_R = 2B(J+1)\ cm^{-1}$

$(J = 1, 2, 3 \ldots\ldots \quad J \neq 0)$

Thus the lines of R brunch would be,

$\bar{\nu}_R = \bar{\nu}_{ex.} \pm 2B(J+1)\ cm^{-1}$, '+' for Antistokes radiation
'−' " stokes radiation.

since $J \neq 0$,

putting $J = 1, 2, 3 \ldots$ we have R brunch lines appear at $2B(1+1) = 4B$, $2B(2+1) = 6B$, $2B(3+1) = 8B$ etc $\ldots\ldots$ cm^{-1}.

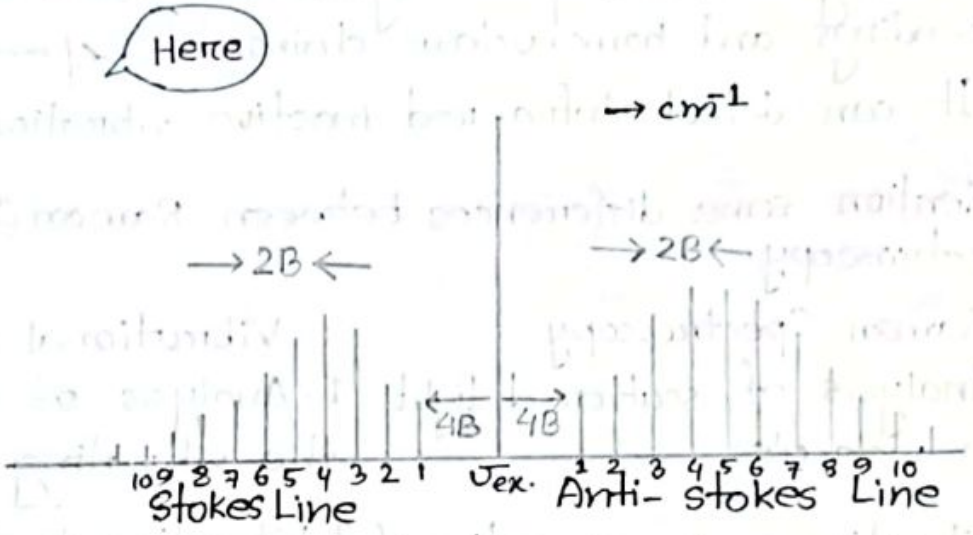


Fig: Lines of R brunch.

□ State the advantages of Raman technique over that of Infra-red in vibrational measurements.

Ans:

The considerable advantages of choosing Raman over infra-red vibrational measurements are:-

a) Water can be used as solvent.
b) Simpler and cheaper instruments than of IR measurements.
c) No sample preparation is required as of IR.
d) Very suitable for biological samples in native state.
e) Has high sensitivity for symmetric molecules, non polare bondings and homonuclear chains.
(f) It can detect infra-red inactive vibration modes.

□ Mention some differences between Raman & vibrational spectroscopy.

Raman Spectroscopy	Vibrational Spectroscopy
1. Analysis of scattered light of molecules.	1. Analysis of absorption of the vibrating molecules.
2. Vibration is Raman active if it causes a change in the polarizability.	2. Vibration is IR active if it causes a change in the dipole moment.
3. Molecules may or may not have dipole moments.	3. Molecules must have an electric dipole moment.
4. Water can be used as solvent	4. Water can not be used as solvent
5. No need of specific sample preparation.	5. Requires specific sample preparation.

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□ What is meant by Raman scattering ?

Ans:

Raman scattering is termed as a phenomenone in which when a electromagnetic radiation with a freequency interacts with a matterr and then scatterred with a freequency higherr orr lowerr than the incident beam.

□ Considerr a symmetricic top molecule $CHCl_3$.

(i) Writte dowrn the equation forr the energy levels of the molecule.

Ans:

The energy expression forr a symmetricic top molecule is,

$$\epsilon_{J,K} = BJ(J+1) + (A-B)K^{-1} \quad cm^{-1}$$

$$(J = 0, 1, 2 \cdots\cdots ; K = \pm J, \pm(J-1) \cdots)$$

(ii) What arre the selection rules forr Raman spectrra in this case?

Ans: The selection rules arre, $\Delta K = 0$

$\Delta J = 0, \pm 1, \pm 2$

[except forr $K=0$ states when $\Delta J = \pm 2$]

(iii) Writte the expression forr the wavenumberr (ν_R and ν_S) equation with mentioning theirr J and ΔJ values at which two serries of R brrunch and S brrunch lines appearr in the Raman spectrra.

Ans:

[R brrunch has been discussed earlierr]

when $\Delta J = +2$, S branch lines are obtained.

$$\Delta \varepsilon_S = \varepsilon_{J+2} - \varepsilon_J$$
$$= B(J+2)(J+2+1) - BJ(J+1)$$
$$= B(J+2)(J+3) - BJ(J+1)$$
$$= B\{J^2 + 2J + 3J + 6 - J^2 - J\}$$
$$= B(4J+6)\ cm^{-1} \quad (J = 0, 1, 2 \cdots)$$

Thus, series of lines obtained in S branch,

$$\bar{\upsilon}_S = \bar{\upsilon}_{ex.} \pm B(4J+6)\ cm^{-1} \quad (J = 0, 1, 2 \cdots)$$

positive sign for antistokes and negative for stokes radiation of S branch.

(iv) Sketch the S branch series only in a figure and show than S branch series occur at 6B, 10B, 14B ... cm^{-1} from the excitation line.

Ans:

From equation of S branch, $\Delta \varepsilon = B(4J+6)\ cm^{-1}$ $(J = 0, 1, 2 \cdots)$

putting $J = 0$, $\Delta \varepsilon_{S=0} = B(0 \times 4 + 6) = 6B\ cm^{-1}$

$J = 1$, $\Delta \varepsilon_{S=1} = B(1 \times 4 + 6) = 10B\ cm^{-1}$

$J = 2$, $\Delta \varepsilon_{S=2} = B(2 \times 4 + 6) = 14\ cm^{-1}$ etc ...

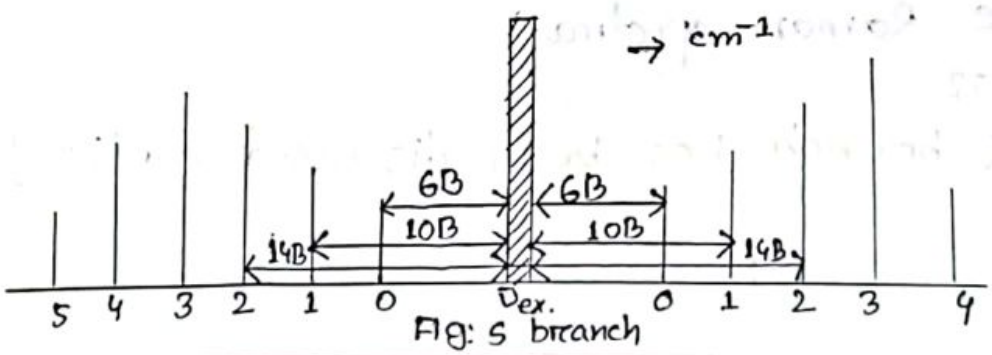


Fig: S branch

□ Which of the vibrations of an AB_3 molecule are infrared or Raman active when it is

(i) trigonal planar (symmetric planar)

(ii) trigonal pyramidal (symmetric pyramidal)?

Ans:

(i) trigonal planar:

- Symmetric stretch (υ_1) → Raman active only
- Out of plane symmetric deformation (υ_2) → IR active
- Asymmetric stretch (υ_3) & asymmetric deformation (υ_4) → Both Raman and IR active.

(ii) Trigonal pyramidal:

All the four vibrational modes (symmetric stretch, out of plane symmetric deformation, asymmetric stretch & asymmetric deformation) are both Raman and IR active.

□ (i) Write down the energy expressions for pure rotational Raman spectra for a linear molecule.

Ans:

The rotational energy levels of linear molecule is,

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \ cm^{-1} \quad (J = 0, 1, 2 \cdots)$$

B = Rotational constant $= \frac{1}{8\pi^2 Ic}$

D = centrifugal distortion

But in Raman spectroscopy, ignoring D, the expression is,

$$\varepsilon_J = BJ(J+1) \ cm^{-1} \quad (J = 0, 1, 2 \cdots)$$

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(ii) Show the rotational energy levels of a diatomic molecule and the rotational Raman spectrum arising from them with showing the relative intensities of lines in the spectrum.

Ans8

The rotational energy level for a linear diatomic molecule

$$\varepsilon_J = BJ(J+1)\ cm^{-1} \quad (J = 0, 1, 2, \ldots)$$

The selection rule,

$$\Delta J = 0 \text{ and } \pm 2 \text{ only,}$$

$\Delta J = 0$ corresponds to the Rayliegh scattering, $\Delta J = \pm 2$ gives rise to the Raman scattering.

We may ignore $\Delta J = -2$ for a pure rotational change.

Thus for $\Delta J = +2$; the energy level we have,

$$\Delta\varepsilon = \varepsilon_{J+2} - \varepsilon_J$$

$$= B(J+2)(J+2+1) - BJ(J+1)$$

$$= B(4J+6)\ cm^{-1} \quad (J = 0, 1, 2, \ldots)$$

Since, $\Delta J = +2$, we may label these lines S branch lines,

$$\Delta\varepsilon_S = B(4J+6)\ cm^{-1}$$

The wavenumbers of the corresponding spectral lines are:

$$\bar{\upsilon}_S = \bar{\upsilon}_{ex.} \pm \Delta\varepsilon_S$$

$$= \bar{\upsilon}_{ex} \pm B(4J+6)\ cm^{-1}$$

S branch series of linear molecules occure at 6B, 10B, 14B..... cm^{-1} from the exciting line.

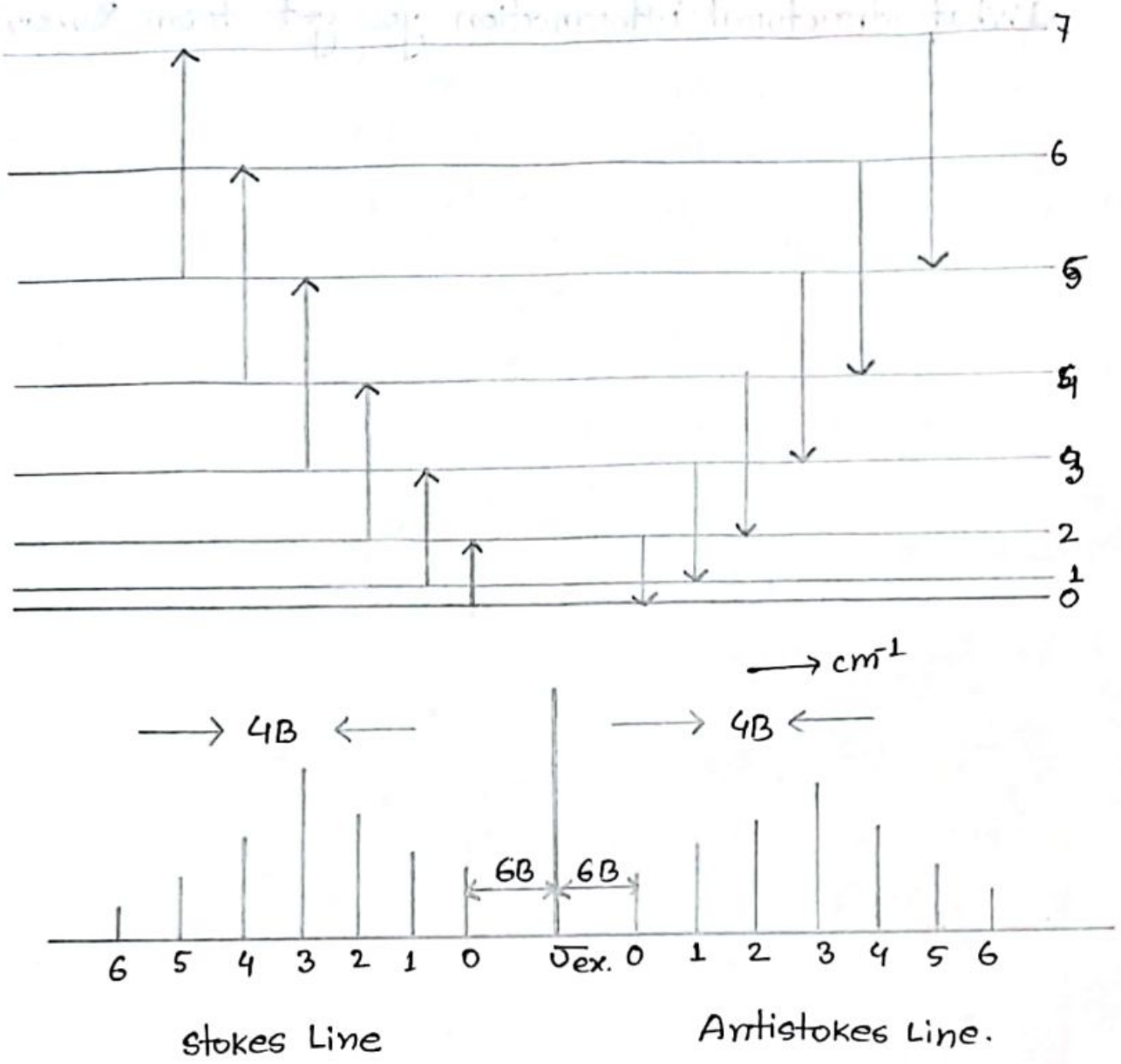


Fig: Rotational energy levels of a diatomic molecule and the rotational Raman spectrum arising from them with showing relative intensities of the lines in the spectrum.

□ What structural information you get from Raman spectrum?

Mathematical Problems:

□ With which type of spectroscopy would one observe the pure rotation spectrum of H_2. ? If the bond length is 0.07417 nm, what would be the spacing of the lines in the spectrum?

We know, $B = \frac{h}{8\pi^2 Ic}$

$= \frac{h}{8\pi^2 \mu r^2 c}$

$= \frac{6.626 \times 10^{-34}}{8\pi^2 \times 8.365 \times 10^{-28} \times (0.07417 \times 10^{-9})^2 \times 3 \times 10^8}$

$= 6078.79\ m^{-1}$

$= 60.79\ cm^{-1}$

Here,

$c = 3 \times 10^8\ ms^{-1}$

$h = 6.626 \times 10^{-34}\ Js$

$\mu = \frac{m_1 m_2}{m_1 + m_2}$

$= \frac{1.673 \times 10^{-27} \times 1.673 \times 10^{-27}}{1.673 \times 10^{-27} + 1.673 \times 10^{-27}}$

$= 8.365 \times 10^{-28}\ kg$

$r = 0.07417\ nm$

$= 0.07417 \times 10^{-9}\ m$

Here the energy spacing will be, $= 4B$

$= (4 \times 60.79)$

$= 243.16\ cm^{-1}$

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□ The wavenumber of the incident radiation in a Raman spectrometer is 20487 cm^{-1}. What is the wavenumber of the scattered stokes radiation for the $J = 2 \leftarrow 0$ transition of $^{14}N_2$? [Given $B = 1.9987\ cm^{-1}$]

We know for S branch,

$$\bar{\upsilon}_S = \bar{\upsilon}_{ex.} \pm B(4J+6)$$

$\bar{\upsilon}_{ex.} = 20487\ cm^{-1}$

$J = 0$

$\bar{\upsilon}_S = ?$

$B = 1.9987\ cm^{-1}$

For stokes radiation,

$$\bar{\upsilon}_S = \bar{\upsilon}_{ex} - B(4J+6)$$

$$= 20487 - 1.9987\ (0+6)$$

$$= 20475\ cm^{-1}$$

[Farzia-50]